IMPLEMENTATION PLAN · OVERVIEW

Size-Level Availability Processor

Nine execution waves and seventeen tasks. Each wave has its own PDF and agent brief in plan/.

9 waves | 17 tasks | Node.js · TypeScript · Express · SQLite

01 02 03 04 05 06 07 08 09

Waves

#	WAVE	TASKS	REQUIRES	UNLOCKS
01	**Foundation** Project skeleton	`scaffold-project`	-	02

02	**Building blocks**
	Validation, retries and storage
		`input-validation`
		`retry-policy`
		`sqlite-schema`
			01	03, 04, 05

03	**Adapters**
	External adapters and persistence
		`product-context-client`
		`version-store`
		`outbox-publisher`
		`availability-client`
			02	04

04	**Core pipeline** Snapshot index and processing pipeline	`snapshot-index` `single-event-pipeline`	02, 03	05, 06
05	**Entry points** Dev route and bulk worker	`dev-messages-endpoint` `bulk-import-worker`	02, 04	06, 07
06	**API and tests** Job status and focused tests	`job-status-endpoint` `focused-tests`	04, 05	07
07	**End-to-end** Verification against WireMock	`e2e-verification`	05, 06	08
08	**Documentation** Solution docs in docs/	`readme-docs`	07	09
09	**Cleanup** Final hygiene	`cleanup-temp`	08	-

Fixed decisions

Stack	Node.js + TypeScript (strict, ESM), Express, better-sqlite3, Vitest.
Per-key serialisation	A lease row in stock_leases inside the database. Not an in-memory lock.
Lease TTL	LEASE_TTL_MS=30000. An expired lease is taken over.
Bulk processing	CONCURRENCY=8 and CHUNK_SIZE=500, both configurable.
Retries	RETRY_MAX_ATTEMPTS=5, RETRY_BASE_MS=200, RETRY_MAX_MS=5000. Exponential backoff with full jitter. Engineering defaults, not from the assignment.

Outbox publishing	Parallel, bounded by CONCURRENCY. Consumers order by sourceVersion.
Documentation	docs/ folder: solution.md, decisions.md, verification.md. The assignment README.md is not modified.
Production broker (documented)	Kafka, partitioned by stock key.

Folder layout

```
plan/
  OVERVIEW.pdf
  wave-01-foundation/   WAVE-01-FOUNDATION.pdf   AGENT.md
  wave-02-building-blocks/   WAVE-02-BUILDING-BLOCKS.pdf   AGENT.md
  wave-03-adapters/   WAVE-03-ADAPTERS.pdf   AGENT.md
  wave-04-pipeline/   WAVE-04-PIPELINE.pdf   AGENT.md
  wave-05-entry-points/   WAVE-05-ENTRY-POINTS.pdf   AGENT.md
  wave-06-api-and-tests/   WAVE-06-API-AND-TESTS.pdf   AGENT.md
  wave-07-e2e/   WAVE-07-E2E.pdf   AGENT.md
  wave-08-documentation/   WAVE-08-DOCUMENTATION.pdf   AGENT.md
  wave-09-cleanup/   WAVE-09-CLEANUP.pdf   AGENT.md
```